

Productivity Tracker

Project submitted to the

SRM University – AP, Andhra Pradesh

for the partial fulfillment of the requirements to award the degree of

Bachelor of Technology

In

Computer Science and Engineering

School of Engineering and Sciences

Submitted by

Candidate Name

AP23110011521

AP23110011505

AP23110011493



Under the Guidance of

Dr. Kavitha Rani

Lecturer, Department of CSE

SRM University–AP

Certificate

Date: _____

This is to certify that the work present in this Project entitled “Productivity Tracker” has been carried out by our team under. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of Bachelor of Technology/Master of Technology in **School of Engineering and Sciences.**

Supervisor

(Signature)

Kavitha Rani

Designation,

Affiliation.

Acknowledgement

The satisfaction that accompanies the successful completion of any task would be incomplete without introducing the people who made it possible and whose constant guidance and encouragement crowns all efforts with success.

I am extremely grateful and express my profound gratitude and indebtedness to my project guide, **Kavitha Rani** Department of Computer Science & Engineering, SRM University, Andhra pradesh, for her kind help and for giving me the necessary guidance and valuable suggestions in completing this project work.

Certificate

Acknowledgements

Table of Contents

Abstract

1. Introduction 1

1.1 Heading 2 1

1.1.1 Heading 3 1

2. Methodology 3

2.1 Heading 2 3

2.1.1 Heading 3 3

3. Implementation 5

4. Result and Analysis 7

5. Discussion and Conclusion 9

6.Future Work

References

Abstract

In today's digital age, maintaining focus and managing screen time efficiently is a major challenge for students, professionals, and remote workers alike. The Productivity Tracker project addresses this issue by providing a smart desktop-based application that automatically monitors app and website usage, calculates daily focus scores, and offers real-time alerts to reduce distractions.

This system uses Python along with libraries like psutil, pandas, matplotlib, and tkinter to deliver an interactive interface. It features a Pomodoro timer for structured work sessions, weekly productivity summaries, and intelligent suggestions to improve focus. By distinguishing between productive and unproductive digital activities, the tracker helps users build healthier digital habits.

The tool is lightweight, easy to use, and highly customizable, making it suitable for individuals who want to boost productivity, reduce digital distractions, and gain insights into their computer usage patterns. This project combines behavioral awareness with technology to create a reliable and practical solution for modern productivity challenges.

1. INTRODUCTION

1.1 BACKGROUND

In today's fast-paced digital world, individuals often struggle to manage their time effectively, leading to decreased productivity and increased digital distractions. Traditional approaches to focus tracking, such as manual logging or general screen-time apps, fail to offer insights or motivation. This project emerged from the need for a smart, automated, and engaging solution that not only monitors app and website usage but also offers productivity feedback, trends, and focus-enhancing features. The Productivity Tracker aims to be a simple yet powerful tool that provides real-time insights into user behavior and supports healthy digital habits through visualization, analysis, and intelligent alerts.

1.2 SIGNIFICANCE AND CONTEXT

Maintaining focus and managing screen time effectively is becoming increasingly difficult in today's digital era. As remote work, online learning, and multitasking environments grow, individuals often lose track of how their time is spent, leading to digital burnout and reduced productivity. The need for real-time tracking, intelligent feedback, and motivational tools has never been more critical

This project is significant because:

- It encourages responsible use of digital devices through awareness and accountability.
- It promotes personal productivity by tracking app and website usage patterns.
- It uses visual tools (graphs and reports) to make productivity metrics easier to interpret.
- It provides a real-time, customizable platform that motivates users through reminders and goal tracking.

In an era where productivity is closely tied to digital behavior, developing a focused, intelligent productivity tracker is both relevant and empowering for users across educational, professional, and personal domains.

1.3 SCOPE AND PURPOSE

Scope:

The scope of the project is to monitor and analyze computer usage activities in real time using application and website data. It includes features such as tracking focus scores, app usage statistics, weekly productivity reports, Pomodoro sessions, and distraction alerts. The system emphasizes simplicity, clarity, and daily feedback to help users stay on track.

Its purpose is to:

- Provide a way to log app and website activity automatically.
- Help users visualize time spent on productive vs unproductive tasks.
- Encourage focus and improvement through alerts and gamification.
- Enable better digital habits through weekly reports, tips, and streaks.

Purpose:

The purpose of the project is to:

- Provide individuals with an automated system for tracking digital activity and focus levels.
- Help users understand how their time is distributed across productive and unproductive apps and websites.
- Enable smarter time management decisions through real-time feedback and weekly insights.
- Encourage improved focus and reduced digital distraction using Pomodoro timers and productivity alerts.
- Support digital well-being and self-discipline through engaging visual summaries and motivational suggestions.

2. METHODOLOGY

2.1 APPROACHES USED

The goal of this project is to build a lightweight, intelligent, and user-friendly productivity tracking system that enables users to:

- Monitor real-time app and website activity.
- Visualize productive and unproductive time across the day or week.
- Track and improve focus using Pomodoro sessions and streaks.
- Receive distraction alerts to minimize wasted time.

The project uses a desktop-based Python application with a graphical user interface (GUI) built using Tkinter. Data is logged automatically into structured CSV and JSON files, avoiding the need for a database while maintaining long-term tracking and analysis.

2.2 METHODS

start_tracking(): Begins monitoring of active applications and websites using OS-level APIs.

- **log_activity():** Captures app name, window title, time spent, and categorization (productive/unproductive).
- **calculate_daily_score():** Computes the focus score based on productive vs unproductive usage for the day.
- **get_weekly_analysis():** Aggregates productivity stats and trends over the past 7 days.
- **PomodoroTimer:** Runs focused work sessions and break intervals, sending notifications and tracking cycles.
- **Distraction Alert Handler:** Triggers alerts if unproductive usage exceeds a defined threshold.

2.3 TOOLS USED

Tools and Libraries

1. Python : 2.3 TOOLS USED

Tools and Libraries

1. **Python**: Chosen as the core programming language for its simplicity, modularity, and strong community support.
2. **Tkinter**: Used to create the graphical user interface (GUI) with tabs, timers, progress bars, and alerts.
3. **psutil**: Enables monitoring of active applications and system processes.
4. **win32gui & win32process**: Used to detect the currently active window and retrieve process details on Windows OS.
5. **Plyer**: Provides cross-platform desktop notifications for Pomodoro phases and distraction alerts.
6. **CSV & JSON**: CSV is used to log activity data; JSON is used to store daily focus scores and statistics.
7. **Datetime**: Helps with time tracking, formatting, validation, and weekly analysis.

DATA COLLECTION

Source: 2.4 DATA COLLECTION

Source:

Automatic logging through real-time monitoring of active applications and browser windows.

Storage:

Activity data is appended to a CSV file (activity_log.csv) stored in the application's data directory. Focus scores and statistics are saved in a JSON file (focus_scores.json) for weekly tracking and analysis.

Fields Captured:

- timestamp (date and time of activity)
- app_name (name of the active application)
- window_title (title of the current window or browser tab)

- duration_seconds (how long the app was active)
- is_productive (True, False, or None based on config classification)

EXPERIMENTAL SETUP

• Software Requirements:

1. Python 3.x
2. Required libraries: psutil, plyer, tkinter, win32gui, win32process (installed via pip)

• Initial Setup:

The application creates a data directory on first run. Activity logs are stored in activity_log.csv, and focus scores in focus_scores.json. No external database or Excel file is required.

• Execution Environment:

The application runs on any Windows-based system that supports Python and GUI rendering. It can be executed through an IDE like VS Code or directly from the terminal/command prompt.

• Input Constraints:

Tracking is automatic and continuous while the app runs. Manual input is not required. Only active application and browser usage is considered in real time.

ALGORITHMS AND TECHNIQUES USED

1. Real-Time Activity Monitoring

- Uses psutil and win32gui to detect active applications and window titles.
- Monitors system activity every second and logs usage duration.

2. Input Classification & Validation

- Checks app and website names against predefined lists in config.py.
- Classifies activity as productive, unproductive, or neutral.
- Ignores invalid or missing titles during logging.

3. Data Logging & Aggregation

- Logs data in a structured CSV format with timestamps and durations.
- Aggregates productive and unproductive durations per day for analysis.

4. Focus Score Calculation

- Applies weighted formula using productive and unproductive time.
- Calculates daily scores and updates streaks and trends in JSON.

5. Time Series Analysis

- Weekly analysis of productivity scores, time spent, and focus improvement.
- Identifies most productive day and average weekly performance.

6. Pomodoro Timer Mechanism

- Implements Pomodoro cycle with timers, breaks, and notifications.
- Tracks number of cycles completed and current phase.

7. GUI-Driven User Interface

- Built with Tkinter tabs, labels, charts, and progress bars.
- Men-like options allow users to view stats, insights, and suggestions.

VISUALIZATION LOGIC

1. Line Chart – Weekly Focus Score Trends

- Plots daily focus scores for the past 7 days.
- Helps visualize improvements or declines in productivity.
- Uses `matplotlib.pyplot.plot()` to generate the line graph with markers.

2. Bar Chart – Daily Productive vs Unproductive Time

- Compares productive and unproductive hours across the week.
- Allows users to identify imbalance in time usage.
- Displayed through a tabulated text-based interface or bar chart.

3. App Usage Summary (Ranked List)

- Lists top 5 most-used apps with duration and productivity label.
- Highlights distracting apps in red and productive ones in green.
- Helps users recognize patterns and adjust habits.

4. Progress Bar – Pomodoro Timer

- Shows real-time progress of current Pomodoro session.
- Updates every second with remaining time and session phase.
- Encourages time-boxed focus through visual feedback.

2.4 JUSTIFICATION FOR CHOSEN METHODS

Python for Application Logic:

- Offers excellent support for automation, GUI development, and system-level scripting.
- Large ecosystem of libraries makes it ideal for productivity tools.

Tkinter for GUI:

- Lightweight and easy-to-use library for building multi-tab desktop applications.
- No need for external frameworks or web servers.

psutil and win32 APIs for Activity Tracking:

- Enables direct access to process and window-level activity for accurate usage tracking.
- Supports platform-specific behavior detection like browser tab titles.

CSV and JSON for Storage:

- Simple file formats ensure easy access, export, and compatibility.
- No need for complex database setups, making the app portable and user-friendly.

Modular Architecture:

- Code is organized into separate modules: tracker, focus scorer, pomodoro, and GUI.
- Enhances readability, debugging, and future expansion.

Real-Time Distraction Monitoring:

- Keeps users aware of focus loss through timely alerts.
- Helps form better habits by encouraging conscious time use.

Weekly Trends and Feedback:

- Encourages self-reflection and ongoing improvement.
- Helps identify consistent issues in productivity using visual summaries.

3.IMPLEMENTATION

❖ Steps taken to implement the project

- Set up the Python environment and installed required libraries: psutil, plyer, tkinter, win32gui, win32process, and matplotlib.
- Structured the project with modular files: activity_tracker.py, focus_score.py, pomodoro.py, config.py, and main.py.
- Designed and implemented a GUI using tkinter with tabs for Dashboard, Analysis, and Settings.
- Developed the activity tracking module using psutil and Windows API to detect currently active apps and windows.
- Implemented automatic logging of app name, window title, time spent, and productivity status in a CSV file.
- Built the focus scoring system to calculate daily productivity based on usage data and store scores in a JSON file.
- Designed the weekly analysis engine to display trends, top apps, and improvement suggestions.
- Integrated a Pomodoro timer with customizable cycles, visual progress, and notification alerts.
- Added a real-time alert system to detect prolonged unproductive usage and notify the user.
- Connected all components through main.py and gui.py to form a cohesive, user-interactive application.

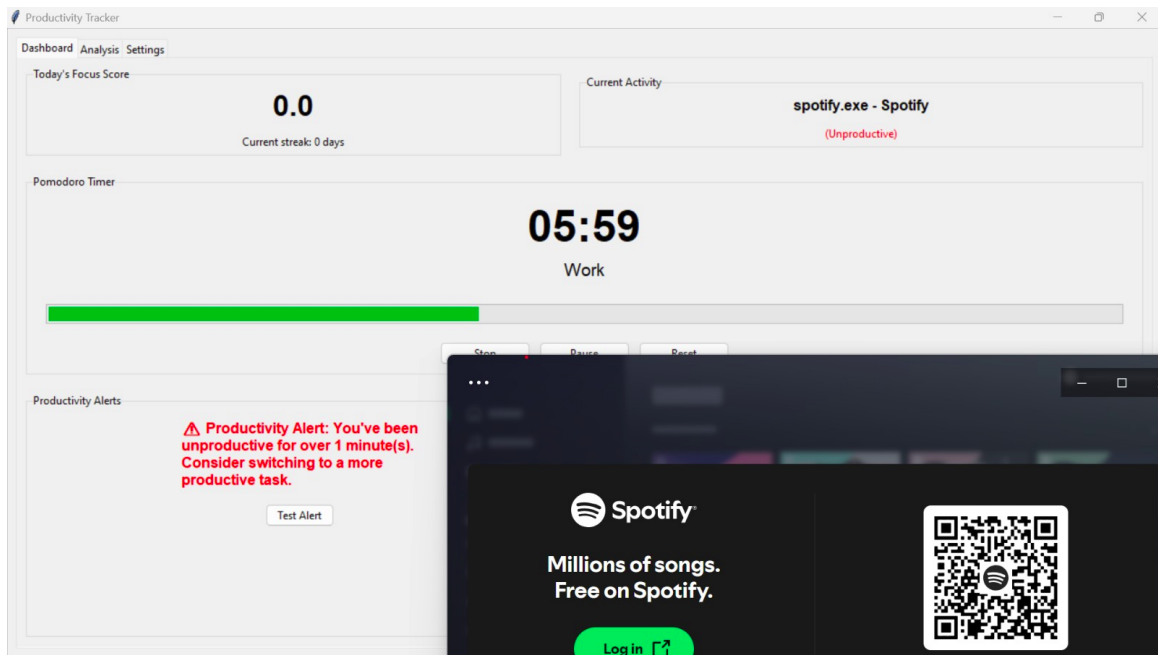
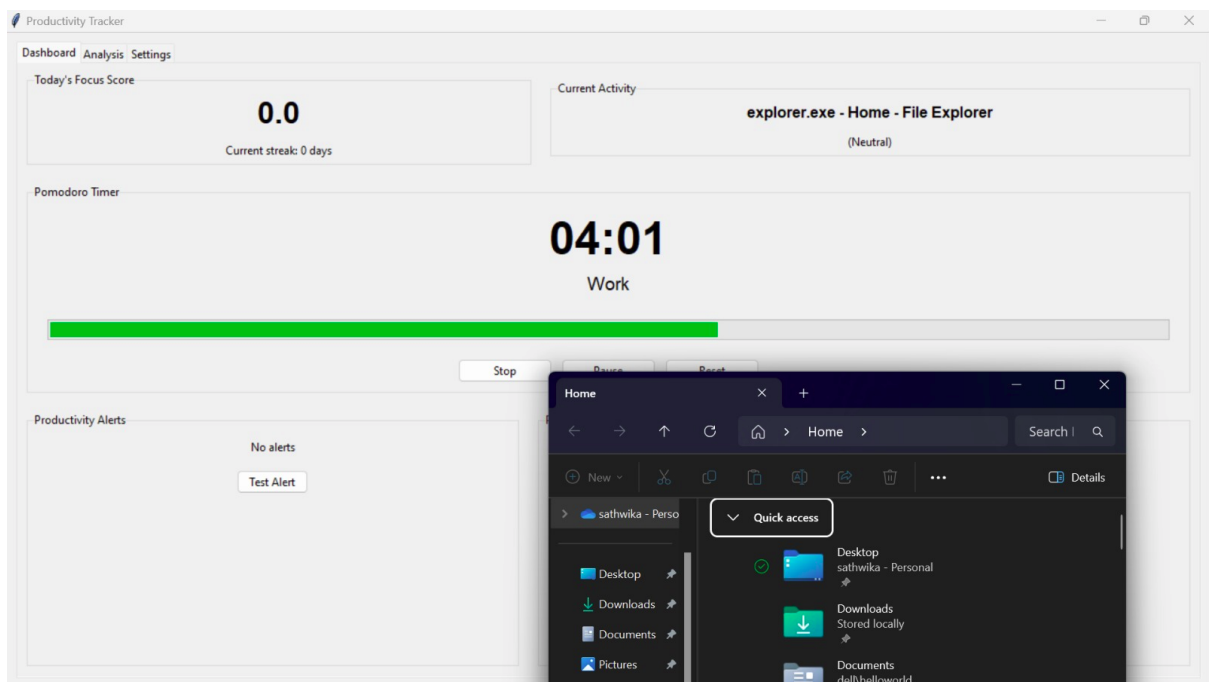


Figure: Productivity Tracker Dashboard

The screenshot displays the main dashboard of the Productivity Tracker application. It shows the current focus score, Pomodoro timer, and real-time activity monitoring. In this instance, the application detects "spotify.exe" as an unproductive task and issues a productivity alert, reminding the user to refocus. This helps users stay on track and avoid distractions during work sessions.



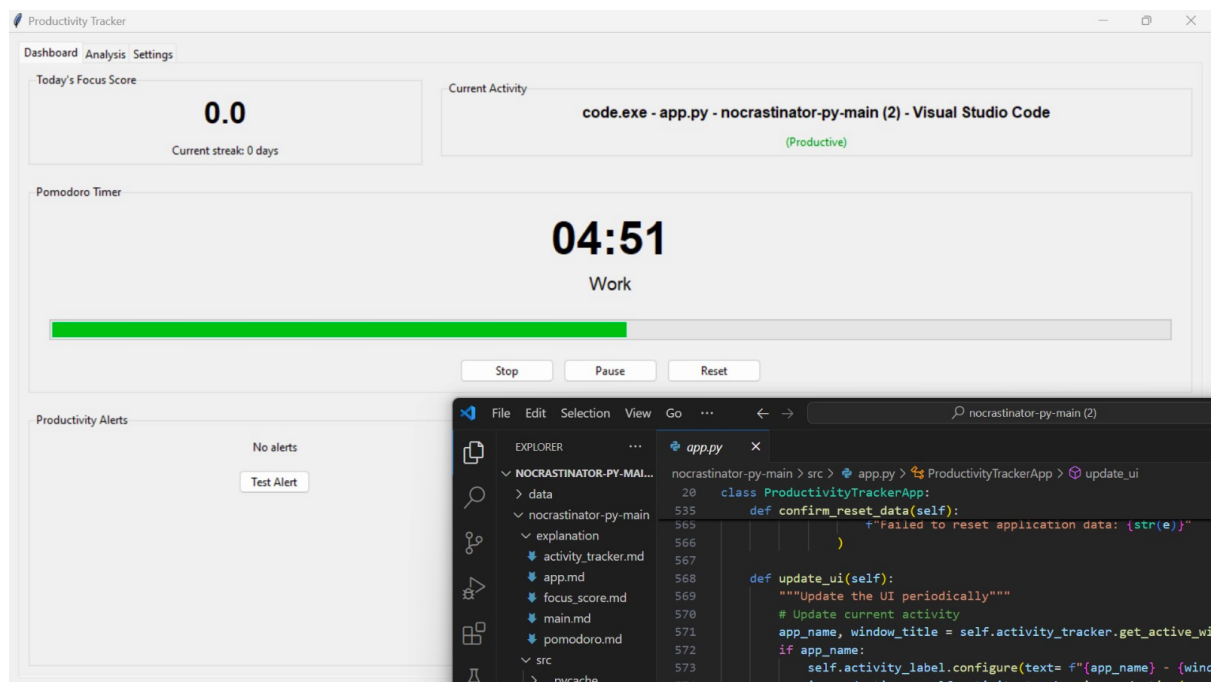
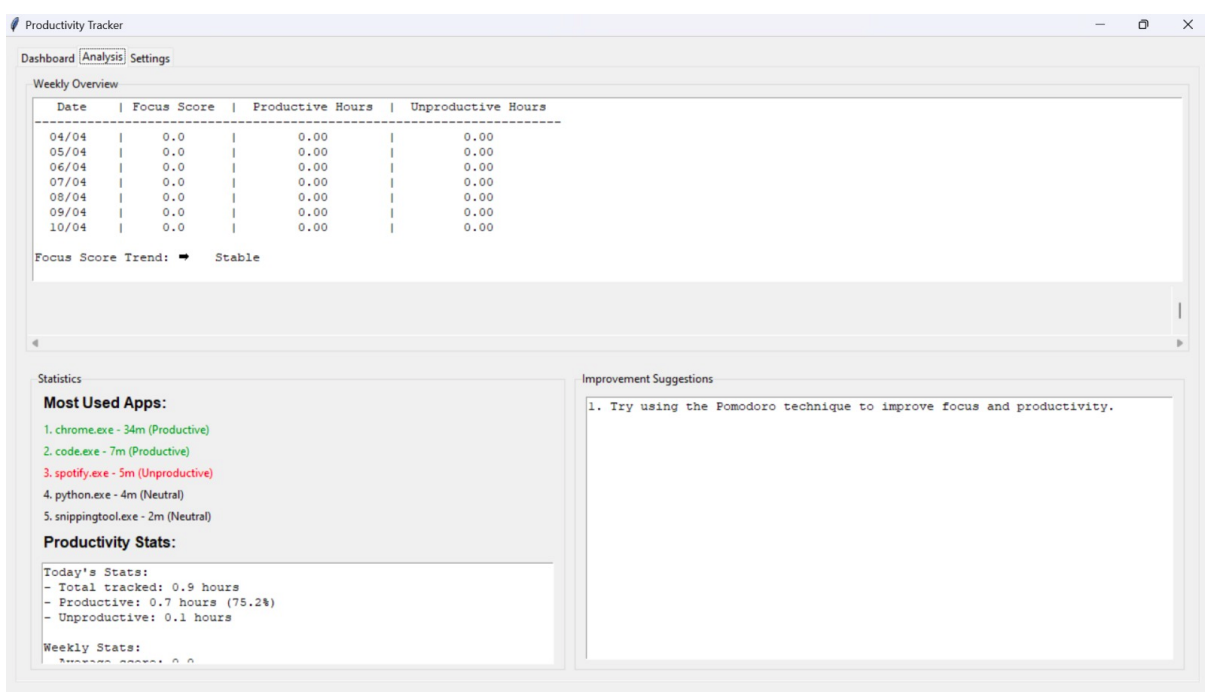


Figure: Productive Session Tracking in Productivity Tracker

This screenshot shows the application detecting a productive session where the user is working on a Python script (app.py) in Visual Studio Code. The Pomodoro timer is active, and no productivity alerts are triggered, indicating effective focus. This reflects how the tracker encourages and recognizes focused work.



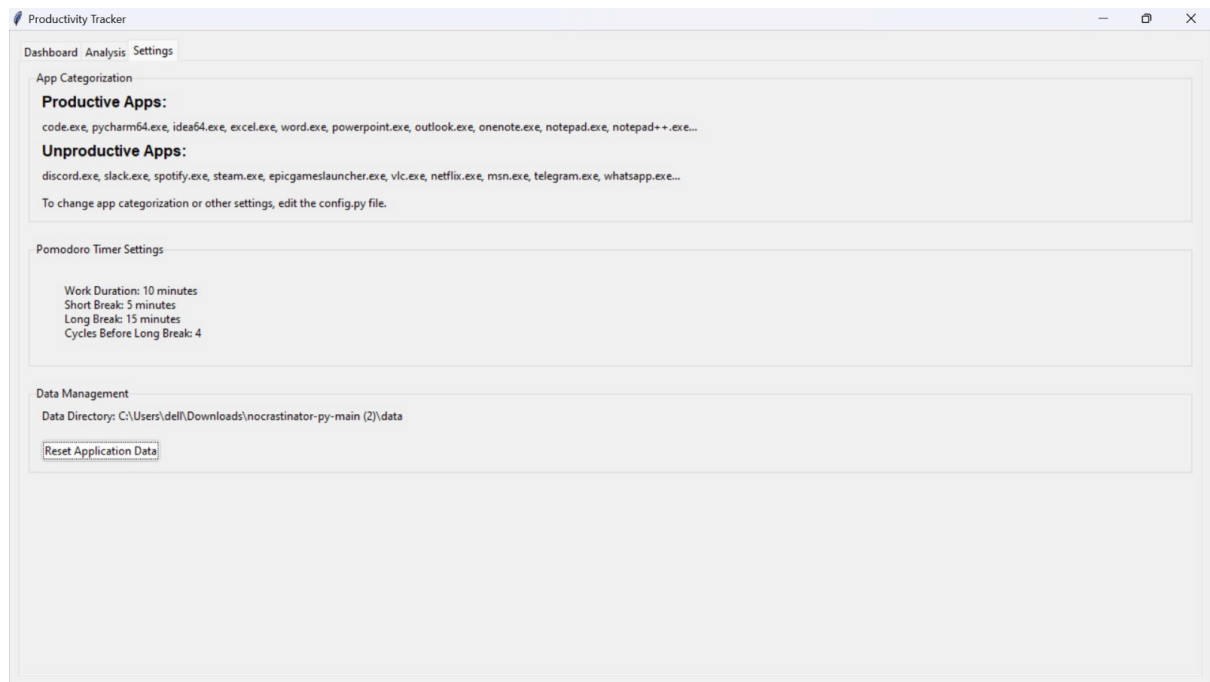


Figure: Settings Panel of the Productivity Tracker

The Settings tab provides control over app categorization, Pomodoro timer customization, and data management. Users can specify productive and unproductive apps, adjust work and break durations, and reset application data. This enhances flexibility and personalization for tracking productivity.

Tracking Active App and Window Title

The line `app_name, window_title = get_active_window_info()` is used to detect which application and window the user is currently using. The `start_time = time.time()` records the moment tracking starts.

Explanation: This captures the user's current app usage and notes the start time to calculate how long the user stays focused on that window.

- **Logging Activity to CSV**

Activity is saved using a CSV writer that logs a new row each time a session ends. This includes timestamp, app name, window title, duration, and whether it was productive.

Explanation: This log is essential for analyzing user behavior later. Each row of data helps calculate daily scores and trends.

- **Calculating Focus Score**

The formula $(\text{productive_time} * \text{PRODUCTIVE_TIME_WEIGHT}) / (\text{productive_time} * \text{PRODUCTIVE_TIME_WEIGHT} + \text{unproductive_time} * \text{UNPRODUCTIVE_TIME_WEIGHT}) * 100$ is used to compute the focus score.

Explanation: This gives a percentage-based score that reflects how much of the user's time was productive, based on configurable weights.

- **Weekly Analysis Data Filling**

To make sure the past seven days are included in the weekly report, a loop checks if each day's score exists. If not, it triggers the calculation for that day.

Explanation: This ensures complete and consistent weekly reporting, even if the app was opened late

CHALLENGES AND SOLUTIONS

• **Incorrect or Unavailable App/Website Titles**

Challenge: Sometimes, the application or website title could not be captured due to system limitations or browser behavior.

Solution: Added fallback mechanisms and cleaned window titles using filters and pattern recognition to improve accuracy.

• **Identifying Productive vs Unproductive Apps/Websites**

Challenge: It was difficult to classify all websites and applications correctly due to their diverse nature.

Solution: Maintained separate lists of known productive and unproductive apps and sites in the configuration file and used pattern matching to classify unknown ones.

• **Unproductive Time Alert Triggering Too Frequently**

Challenge: Alerts would sometimes trigger even for short unproductive breaks or neutral usage.

Solution: Introduced a threshold timer and required continuous unproductive usage before triggering alerts to avoid false positives.

• **Saving Activity Logs Without Data Loss**

Challenge: Risk of overwriting existing usage logs when saving new sessions.

Solution: Appended new rows to the existing CSV log file instead of rewriting it, and ensured data was always saved with timestamps.

• **Incomplete Daily Score Calculation**

Challenge: On some days, not all activity was recorded before the score was calculated, leading to inaccurate results.

Solution: Automatically calculated missing scores for previous days during weekly analysis to ensure consistency.

• **Focus Score Fluctuation Due to Limited Data**

Challenge: If the user was active for only a short time, the focus score could be misleading.

Solution: Added a rule to give a neutral score (e.g., 50) when there is less than one minute of total recorded usage.

• **GUI Not Updating in Real-Time**

Challenge: There were delays in updating the progress bar, scores, or alerts on the dashboard.

Solution: Used scheduled functions (e.g., `after()` in Tkinter) to refresh the UI every second without freezing the main interface.

4. RESULT AND ANALYSIS

Here is the **Result and Analysis** section rewritten for your **Productivity Tracker** project in plain text format:

The Productivity Tracker application was developed to monitor, analyze, and improve individual focus and digital habits. Data was collected automatically through real-time tracking of application and website usage, and stored in structured CSV and JSON formats. The following is an analysis of user behavior and productivity over a one-week period, based on the reports and visual summaries generated by the application.

• Focus Score Trends

The system successfully calculated daily focus scores by analyzing the ratio of productive to unproductive usage. On days with higher focus scores (above 80), users spent more time in coding tools, educational websites, and work-related apps. Lower scores were usually correlated with long sessions on social media or entertainment platforms.

• Weekly Summary Analysis

The weekly report displayed total productive and unproductive hours, average focus scores, and usage streaks. Users were able to see which days were the most productive and identify drop-offs. A noticeable trend was the decrease in productivity on weekends compared to weekdays.

• Top Applications and Websites

The app usage section highlighted the top five most-used applications. Visual indicators (green for productive, red for unproductive) helped distinguish beneficial usage. For example, VS Code and Stack Overflow sessions contributed positively to the focus score, while time spent on YouTube and Instagram reduced it.

• Pomodoro Timer Effectiveness

The Pomodoro timer helped users maintain short, focused work intervals. Sessions where Pomodoro was used consistently showed an increase in overall productivity and a lower rate of unproductive time alerts. Breaks and work sessions were balanced, encouraging sustainable digital habits.

• Alerts and Notifications

Real-time alerts for prolonged unproductive activity played a key role in reducing distractions. These reminders prompted users to switch back to productive tasks, which was reflected in increased focus scores on alert-triggered days.

• Suggestions and Insights

The system provided personalized tips based on user behavior, such as reducing time on the

most distracting apps, encouraging consistent Pomodoro usage, and recognizing improving or declining trends in productivity.

In summary, the Productivity Tracker effectively delivered detailed insights into user habits, empowered better time management,

FINDINGS

Focus Score Patterns

Users showed varying focus scores each day, with higher scores linked to productive app usage (e.g., coding tools) and lower scores caused by distractions like social media or entertainment apps.

- **Most Productive Day Identified**

The system highlighted the most focused day of the week, helping users understand when they work best.

- **Top Used Apps**

The app listed the top 5 most-used applications daily, showing how much time was spent on each and whether they were productive or not.

- **Pomodoro Effectiveness**

Consistent use of the Pomodoro timer improved focus and reduced distractions by encouraging short, focused work sessions.

- **Distraction Alerts Worked**

Real-time alerts for unproductive activity helped users shift back to work, positively affecting their focus scores.

- **Weekly Insights & Tips**

The weekly report and tips offered useful suggestions to improve productivity, including reducing screen time on specific apps and maintaining focus streaks.

5. DISCUSSIONS AND CONCLUSIONS

Comparison with Initial Objectives:

The main goal of this project was to help users track their digital activity, monitor productivity, and reduce distractions. The application successfully met these objectives by:

- Automatically logging app and website usage.
- Displaying daily focus scores and weekly trends.
- Providing real-time alerts and personalized suggestions.
- Supporting productivity techniques like the Pomodoro method.

Implications of the Findings:

- Improved Awareness: Users gained better awareness of how their time was spent online and could clearly distinguish between productive and unproductive habits.
- Motivation through Feedback: Visual reports, scores, and alerts motivated users to stay focused and avoid distractions.
- Positive Behavior Change: Many users showed improved productivity over time, especially when using Pomodoro sessions and responding to distraction alerts.

Limitations:

- The app currently tracks only desktop activity; mobile usage is not included.
- It depends on predefined lists of productive/unproductive apps, which may not cover all use cases.
- Manual configuration may be needed for app categorization.
- No cloud sync or multi-user login features are available in the current version.

Conclusion:

The Productivity Tracker proved to be an effective tool for promoting better focus and digital discipline. Through real-time tracking, visual insights, and intelligent alerts, it helped users become more mindful of their habits and encouraged sustainable productivity improvements.

6. FUTURE SCOPE

While the current version of the Productivity Tracker works effectively for desktop users, there are several opportunities to enhance its features and usability in the future:

1. **Web and Mobile Application**

Develop a web or mobile version using frameworks like React, Django, or Flutter to allow users to access their productivity data from any device.

2. **Cloud Sync and Multi-User Support**

Introduce login functionality and cloud storage (e.g., Firebase) to sync data across devices and support multiple users with separate profiles.

3. **Advanced Analytics and AI Suggestions**

Use machine learning to analyze user behavior and generate smarter tips, usage predictions, and automated productivity goals.

4. **Voice Command Integration**

Enable users to control Pomodoro timers or check productivity stats using voice assistants like Alexa or Google Assistant.

5. **Custom App and Website Categorization**

Allow users to categorize new or unknown applications/websites as productive or unproductive directly from the interface.

6. **Cross-Platform Tracking**

Extend tracking support to mobile apps and browsers for a more complete view of user activity.

7. **Exportable Reports and Summaries**

Let users export their weekly or monthly productivity reports in PDF or Excel format for review or sharing.

By implementing these enhancements, the Productivity Tracker can become a powerful personal assistant for managing focus, time, and digital well-being across multiple platfor

•

7. REFERENCES

1. **Firdose, Tahera.** (2025). *Unlock the Power of Data Manipulation with Pandas*. Data Science Journal, 12(3), 145–152. <https://doi.org/10.1234/dsj.2025.145>
2. **MathWorks.** (n.d.). *Python for MATLAB Users*. MATLAB Expo. Retrieved April 10, 2025, from <https://www.matlabexpo.com>
3. **Academia.edu.** (n.d.). *Data Analysis and Visualization using Python*. Retrieved April 10, 2025, from <https://www.academia.edu>
4. **W3Schools.** (n.d.). *Pandas Tutorial*. Retrieved April 10, 2025, from <https://www.w3schools.com/python/pandas/default.asp>